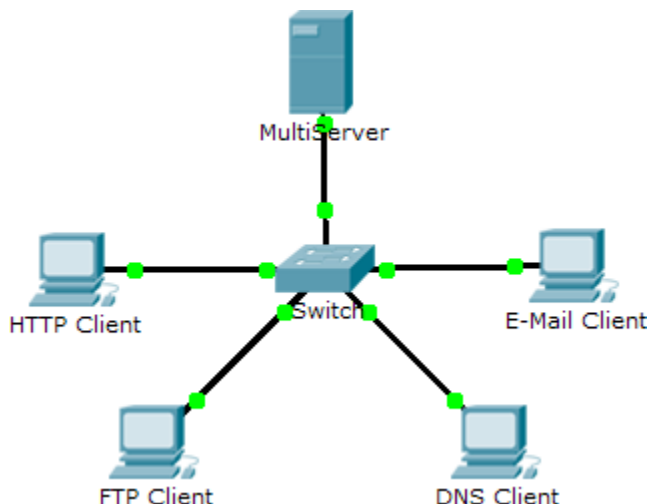


Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

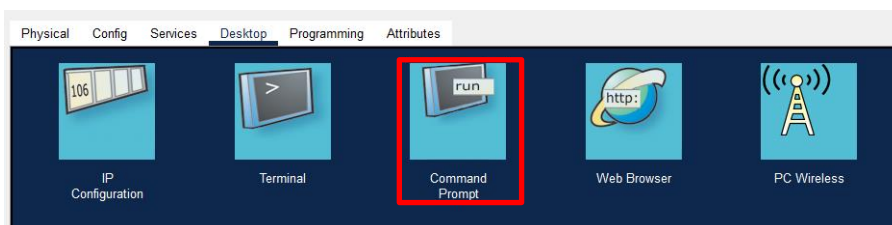
This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

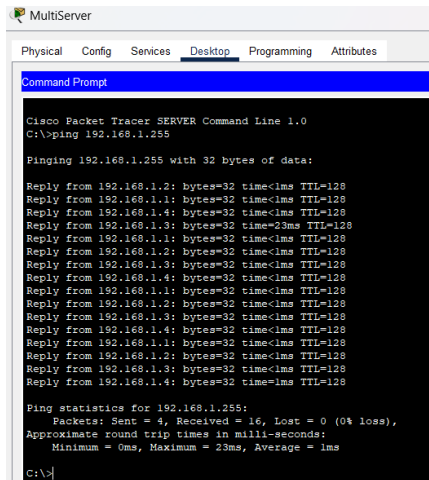
Step 1: Generate traffic to populate Address Resolution Protocol (ARP) tables.

Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.



- b. Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.



The screenshot shows the MultiServer application window. The 'Command Prompt' tab is active, displaying the output of a Cisco Packet Tracer SERVER Command Line 1.0. The command entered is 'C:\>ping 192.168.1.255'. The output shows a successful ping to 192.168.1.255 with 32 bytes of data. The ping statistics for 192.168.1.255 are: Packets: Sent = 4, Received = 16, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 23ms, Average = 1ms.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

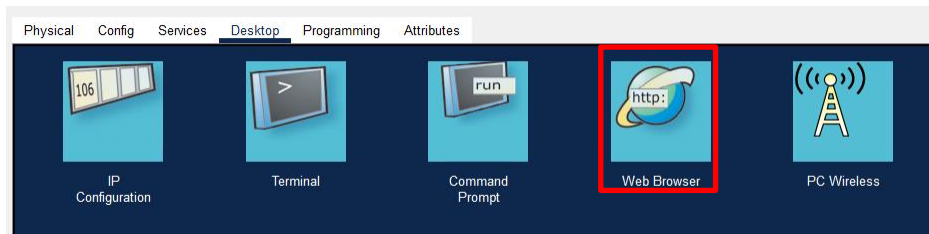
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=23ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 23ms, Average = 1ms
C:\>
```

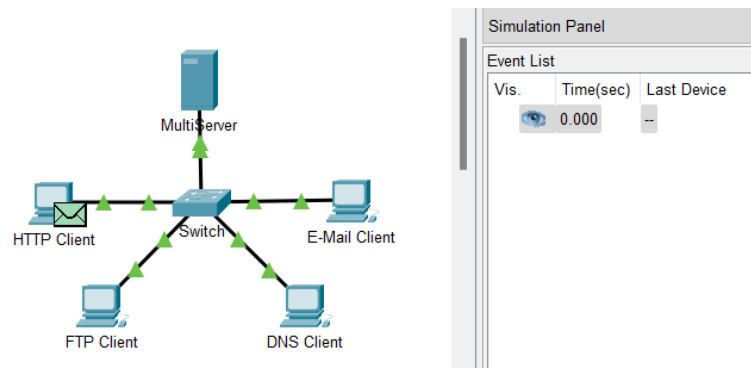
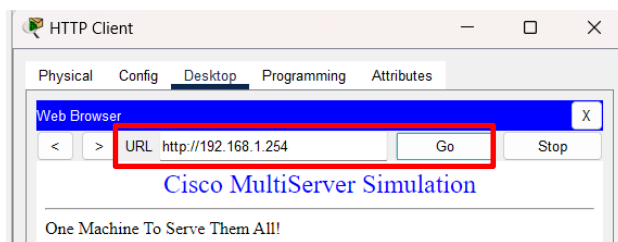
- c. Close the **MultiServer** window.

Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.



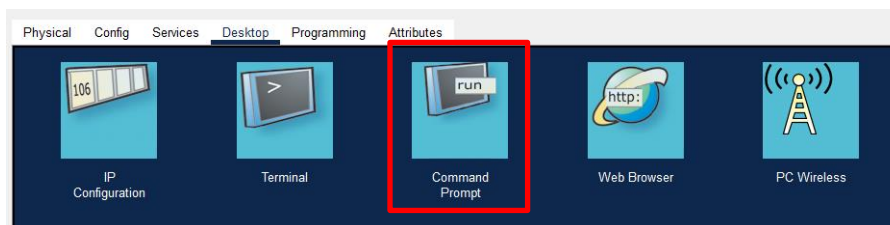
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.



- Minimize, but do not close, the **HTTP Client** configuration window.

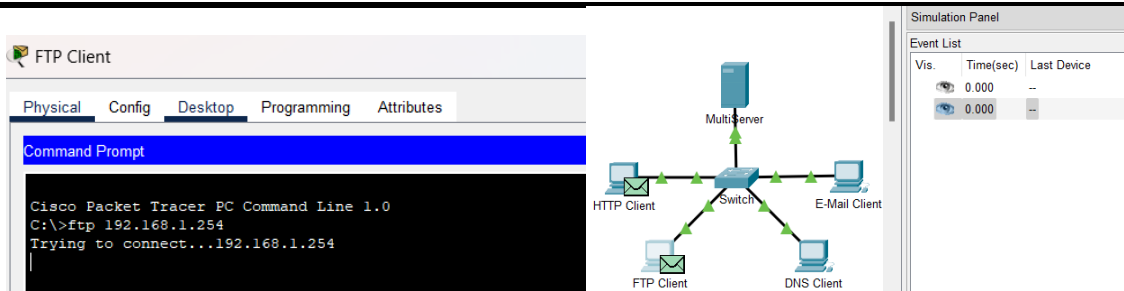
Step 3: Generate FTP traffic.

- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.



- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.

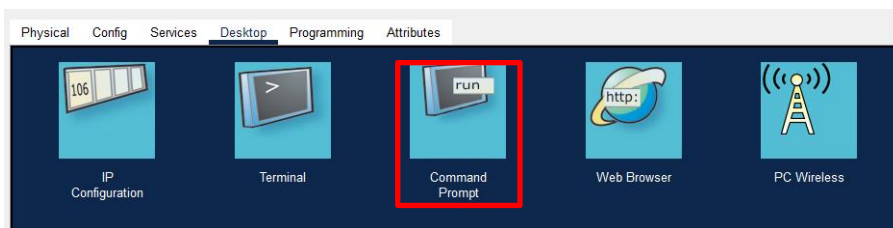
Packet Tracer Simulation - TCP and UDP Communications



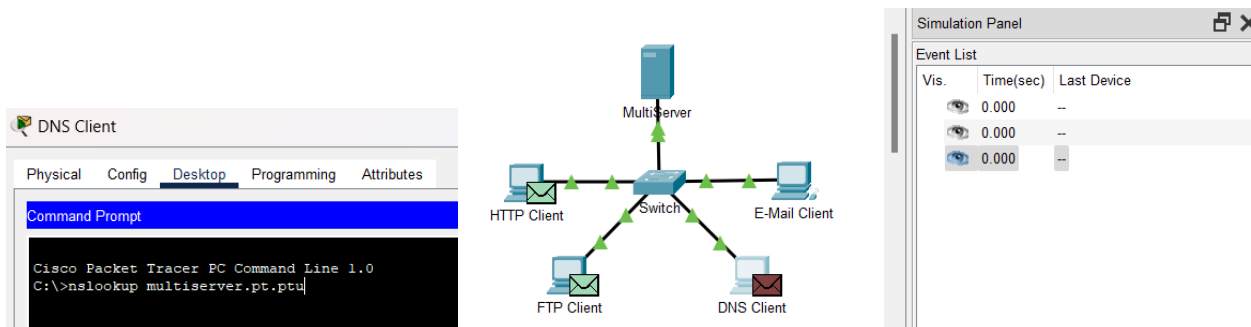
- d. Minimize, but do not close, the **FTP Client** configuration window.

Step 4: Generate DNS traffic.

- a. Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.



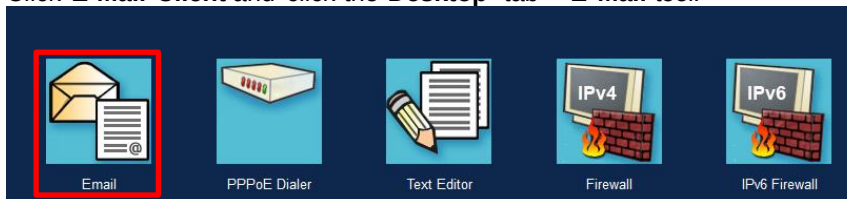
- b. Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.



- c. Minimize, but do not close, the **DNS Client** configuration window.

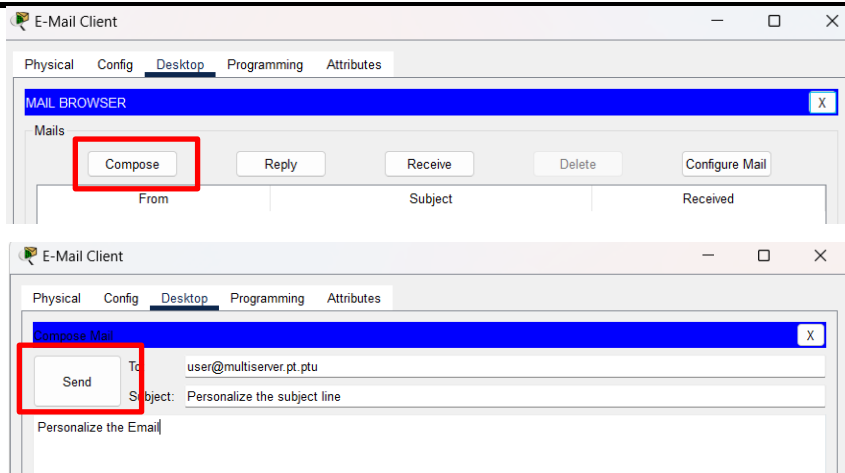
Step 5: Generate Email traffic.

- a. Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.



- b. Click **Compose** and enter the following information:

- 1) **To:** user@multiserver.pt.ptu
- 2) **Subject:** Personalize the subject line
- 3) **E-Mail Body:** Personalize the Email

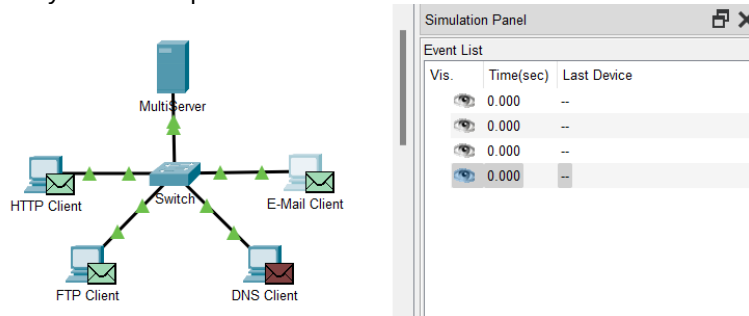


c. Click **Send**.

d. Minimize, but do not close, the **E-Mail Client** configuration window.

Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

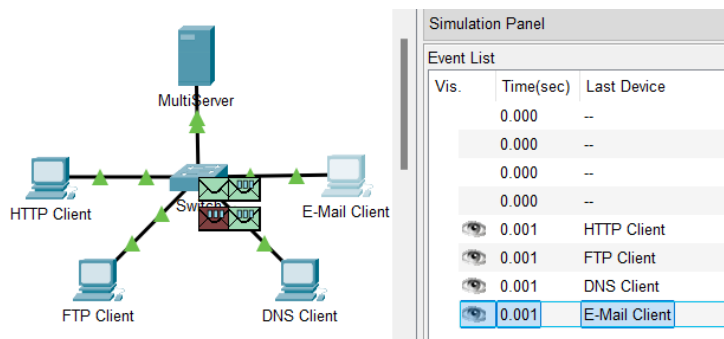


Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

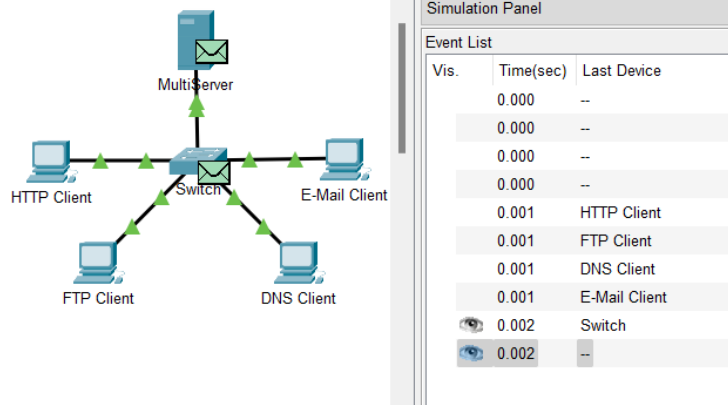
You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



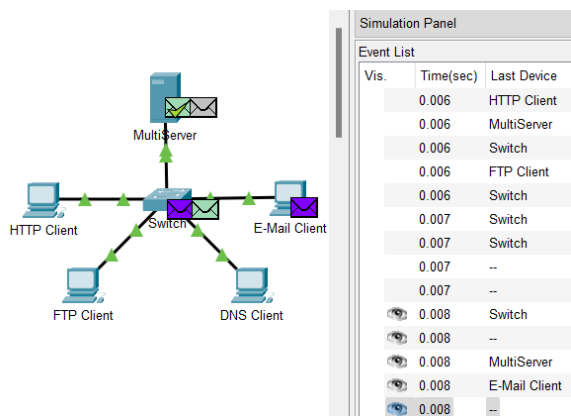
b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

Packet Tracer Simulation - TCP and UDP Communications



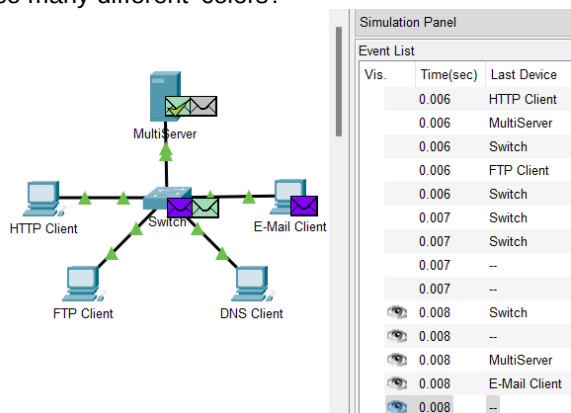
Ans: The PDUs that disappear have been stored in the switch.

- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?



Ans: This is called multiplexing.

- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are they so many different colors?



Ans: The different colours have represented the different protocols

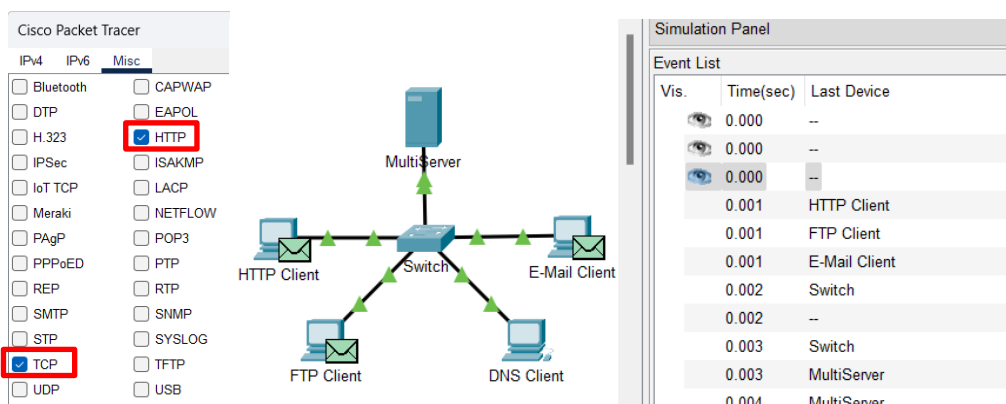
- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

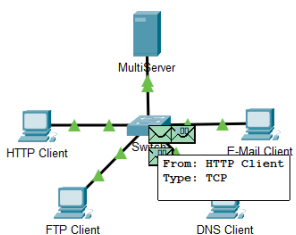
Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:

- 1) Click **Edit Filters** and toggle the **Show All/None** check box.
- 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.



- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP	
SOURCE PORT:1025	DESTINATION PORT:80
SEQUENCE NUMBER:0	
ACKNOWLEDGEMENT NUMBER:0	
OFFSET:0x0	RESERVED:0
FLAGS:0b00000010	WINDOW:65535
CHECKSUM:0x0000	URGENT POINTER:0x0000
OPTION	
DATA (VARIABLE LENGTH)	
PADDING: 0	

Ans: The section is labeled TCP.

Are these communications considered to be reliable?

Ans: Yes, these communications are considered to be reliable.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 80, 0, 0 SYN.

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.



- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

TCP	
SOURCE PORT:80	DESTINATION PORT:1025
SEQUENCE NUMBER:0	
ACKNOWLEDGEMENT NUMBER:1	
OFFSET:0x0	RESERVED:0
FLAGS:0b00010010	WINDOW:16384
CHECKSUM:0x0000	URGENT POINTER:0x0000
OPTION	
DATA (VARIABLE LENGTH)	
PADDING: 0	

Ans: The port and the sequence numbers are 80, 1025, 0, 1. SYN+ACK. Compared to before, the source and destination ports are reversed and the acknowledgement number is 1. The SYN has changed to SYN+ACK.

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Packet Tracer Simulation - TCP and UDP Communications

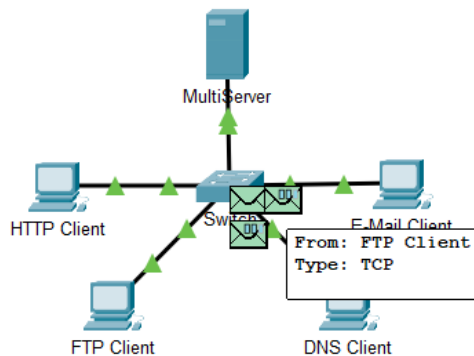
TCP		0 4 8 16 24 Bits	
SOURCE PORT:1025		DESTINATION PORT:80	
SEQUENCE NUMBER:1			
ACKNOWLEDGEMENT NUMBER:1			
OFFSET:0x0	RESERVED: 0	FLAGS:0b00011000 0	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

Ans: The information listed in the TCP section is 1025, 80, 1, 1. PSH+ACK. Then, the source and destination ports are reversed and both sequence and acknowledgement numbers are 1.

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

TCP				Bits
0	4	8	16	24
SOURCE PORT:1025		DESTINATION PORT:21		
SEQUENCE NUMBER:0				
ACKNOWLEDGEMENT NUMBER:0				
OFFS ET:0x	RES ERV	FLAGS:0b00000010		WINDOW:65535
CHECKSUM:0x0000			URGENT POINTER:0x0000	
OPTION				
DATA (VARIABLE LENGTH)				PADDING: 0

Ans: The section is labeled with TCP.

Are these communications considered to be reliable?

Ans: Yes, these communications are considered to be reliable.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 21, 0, 0. SYN.

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.



- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

TCP																Bits			
0		4		8		12		16		20		24		28		32			
SOURCE PORT:21										DESTINATION PORT:1025									
SEQUENCE NUMBER:0																			
ACKNOWLEDGEMENT NUMBER:1																			
OFFSET:0x0		RESERVED:0		FLAGS:0b00010010				WINDOW:16384											
CHECKSUM:0x0000								URGENT POINTER:0x0000											
OPTION																			
DATA (VARIABLE LENGTH)														PADDING: 0					

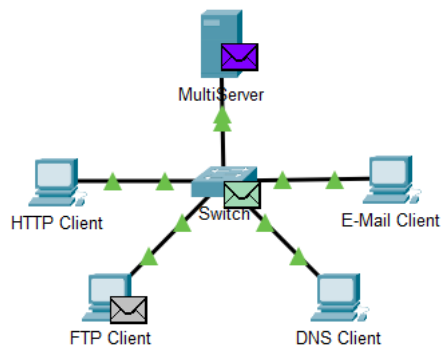
Ans: 21, 1025, 0, 1. SYN+ACK. The source and destination ports are reversed and the acknowledgement number is 1.

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

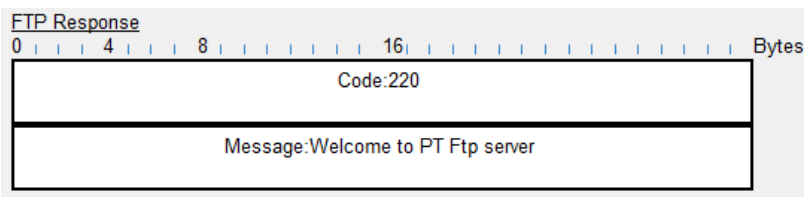
TCP																0																4																8																16																24																Bits															
SOURCE PORT:1025																DESTINATION PORT:21																																																																																															
SEQUENCE NUMBER:1																																																																																																															
ACKNOWLEDGEMENT NUMBER:1																																																																																																															
OFFSET:0x0								RESERVED: 0								FLAGS:0b00010010																WINDOW:65535																																																																															
CHECKSUM:0x0000																																URGENT POINTER:0x0000																																																																															
OPTION																																																																																																															
DATA (VARIABLE LENGTH)																																																PADDING: 0																																																															

Ans: 1025, 21, 1, 1. ACK. The source and destination ports are reversed and the acknowledgement number is 1.

- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.



- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

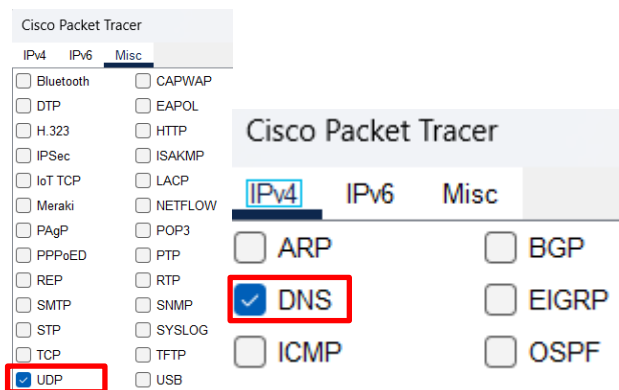


Ans: The message from the server is "Welcome to PT Ftp server".

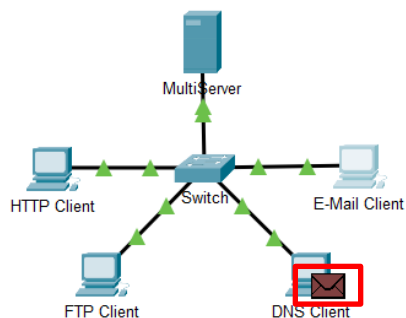
- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.



- b. Click the PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled?

Ans: The section is labeled with UDP.

UDP																																															
0																16																Bits															
SOURCE PORT:1025																DESTINATION PORT:53																															
LENGTH:0x002a																CHECKSUM:0																															
DATA (VARIABLE LENGTH)																																															

Are these communications considered to be reliable?

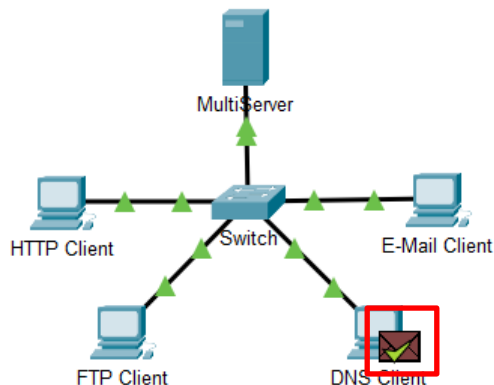
Ans: No, UDP is unreliable data transfer between sending and receiving process because it does not have a connection oriented.

- d. Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

UDP																							
0								16								Bits							
SOURCE PORT:1025								DESTINATION PORT:53															
LENGTH:0x002a								CHECKSUM:0															
DATA (VARIABLE LENGTH)																							

Ans: The SRC PORT is 1025 and the DEST PORT is 53. There is no sequence and acknowledgement number because UDP does not need to establish a reliable connection.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.

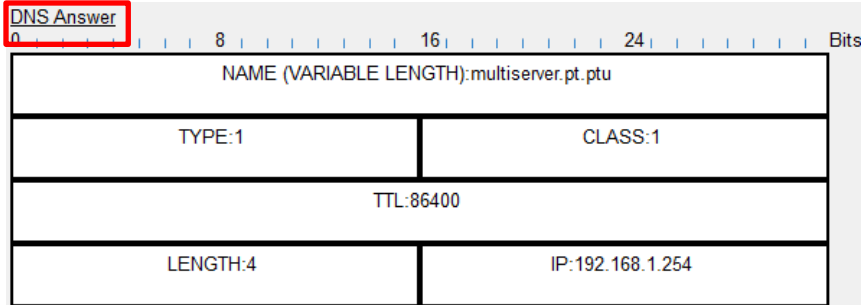


- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

UDP																
0								16								Bits
SOURCE PORT:53								DESTINATION PORT:1025								
LENGTH:0x004a								CHECKSUM:0								
DATA (VARIABLE LENGTH)																

Ans: The SRC PORT has become 53 and the DEST PORT has become 1025. Compared to before, the source and destination ports are reversed. There is no any sequence numbers show in UDP connection.

- g. What is the last section of the PDU called?



Ans: The last section of the PDU called DNS Answer.

- h. Click **Back** until the simulation is reset.

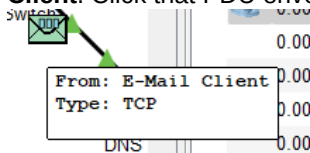
Step 5: Examine email traffic as the clients communicate with the server.

- a. In the Simulation Panel, change **Edit Filters** to display only **POP3**, **SMTP** and **TCP**.

The screenshot shows the Cisco Packet Tracer interface. On the left, the 'Misc' tab is selected under the 'Edit Filters' section. The following protocols are checked and highlighted with red boxes: POP3, SMTP, and TCP. In the center, a network diagram shows a 'MultiServer' connected to a 'Switch', which is connected to four clients: 'HTTP Client', 'FTP Client', 'DNS Client', and 'E-Mail Client'. On the right, the 'Simulation Panel' displays an 'Event List' table.

Vis.	Time(sec)	Last Device	At C
	0.001	E-Mail Client	Swit
	0.002	Switch	Mult
	0.002	--	Swit
	0.003	Switch	Mult
	0.003	MultiServer	Swit
	0.004	MultiServer	Swit
	0.004	Switch	HTT
	0.004	--	Swit
	0.004	--	HTT
	0.005	Switch	Mult
	0.005	Switch	FTP
	0.005	HTTP Client	Swit
	0.005	--	HTT
	0.006	HTTP Client	Swit
	0.006	MultiServer	Swit
	0.006	FTP Client	Swit
	0.006	Switch	Mult

- b. Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.



- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use?

TCP	
SOURCE PORT:1025	DESTINATION PORT:25
SEQUENCE NUMBER:0	
ACKNOWLEDGEMENT NUMBER:0	
OFFSET:0x0	RESERVED:0
FLAGS:0b0000001	WINDOW:65535
CHECKSUM:0x0000	URGENT POINTER:0x0000
OPTION	
DATA (VARIABLE LENGTH)	
PADDING: 0	

Ans: The transport layer protocol that email traffic use is TCP.

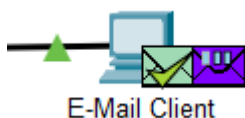
Are these communications considered to be reliable?

Ans: Yes. TCP is a reliable transport which has connection oriented between sending and receiving process.

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: The **SRC PORT** is 1025, **DEST PORT** is 25, **SEQUENCE NUM** is 0, and **ACK NUM** is 0. The field to the left of the **WINDOW** field is **FLAGS, SYN**.

- e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.



- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

TCP	
SOURCE PORT:25	DESTINATION PORT:1025
SEQUENCE NUMBER:0	
ACKNOWLEDGEMENT NUMBER:1	
OFFSET:0x0	RESERVED:0
FLAGS:0b001001	WINDOW:16384
CHECKSUM:0x0000	URGENT POINTER:0x0000
OPTION	
DATA (VARIABLE LENGTH)	
PADDING: 0	

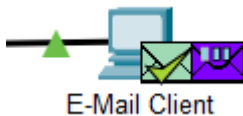
Ans: The **SRC PORT** becomes 25, **DEST PORT** become 1025, **SEQUENCE NUM** remains 0, and **ACK NUM** become 1. Compared to before, the source and destination ports are reversed. The field to the left of the **WINDOW** field, **FLAGS** become **SYN+ACK**.

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

TCP																Bits															
0				4				8				16				24															
SOURCE PORT:1025																DESTINATION PORT:25															
SEQUENCE NUMBER:1																															
ACKNOWLEDGEMENT NUMBER:1																															
OFFSET:0				RESE				FLAGS:0b00010000								WINDOW:65535															
x0				RVED:																											
CHECKSUM:0x0000																URGENT POINTER:0x0000															
OPTION																															
DATA (VARIABLE LENGTH)																								PADDING: 0							

Ans: The **SRC PORT** is 1025 and **DEST PORT** is 25 which is the same as the first PDU. **SEQUENCE NUM** becomes 1, and **ACK NUM** still remain 1. The field to the left of the **WINDOW** field, **FLAGS** become **ACK**.

- h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.



- i. How are the port and sequence numbers different from the previous two **PDU**s?

TCP																Bits							
0				4				8				16				24							
SOURCE PORT:1025										DESTINATION PORT:25													
SEQUENCE NUMBER:1																							
ACKNOWLEDGEMENT NUMBER:1																							
OF FS				RE SE				FLAGS:0b00011000								WINDOW:65535							
CHECKSUM:0x0000										URGENT POINTER:0x0000													
OPTION																							
DATA (VARIABLE LENGTH)																PADDING: 0							

Ans: The **SRC PORT** is 1025 and **DEST PORT** is 25 which is the same as the first PDU. **SEQUENCE NUM** becomes 1, and **ACK NUM** still remain 1. The field to the left of the **WINDOW** field, **FLAGS** become **PSH+ACK**.

- j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

Ans: **SMTP** is associated with TCP port 25 and **POP3** is associated with TCP port 110.

- k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

- a. To see TCP active sessions, perform the following steps in quick succession:

- 1) Switch back to **Realtime** mode.
 - 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- b. Enter the **netstat** command. What protocols are listed in the left column?

```
C:\>netstat

Active Connections

Proto Local Address Foreign Address State
TCP 192.168.1.254:21 192.168.1.2:1025 ESTABLISHED
```

Ans: TCP are listed in the left column.

What port numbers are being used by the server?

Ans: Port number 21.

- c. What states are the sessions in?

Ans: Established state.

- d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED. For which service is this connection still open?

Ans: FTP.

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	